

Evaluating Data

paranoia is a ~~valuable~~ necessary trait

poking holes is job one

- **Do Not Trust Your Data.**
 - even – especially! - if you made it.
- data comes from people. people make mistakes.
- first law of data entry: the more you type, the more mistakes you make.

what to look for

- missing values
- inconsistencies between data points
- out-of-range values
- suspicious repeats / duplicates
- misspellings / typos
- truncations within cells
- entire rows truncated from your dataset

Many of these are in the DC 311 data.

truncations — within cells, and whole rows



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Just received an Excel file that's exactly
65,536 rows 🚨

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- $65,536 = 2^{16}$, the old Excel row ceiling. an exact power of two is a red flag that rows fell off the bottom.
- our file is 9,153 rows — a slice, not the firehose.
- *always ask: what got left out before this reached me?*

twitter.com/bendystraw/status/736270952670081026

start evaluating your data

- first, the basic question: what is this?
- then, sort and filter to get familiar
- next, pivot tables to see distinct values

every step informs one judgment: can this data actually answer my question?

pivot tables

- great for when you need to see distinct values
- distinct means exactly that — any difference makes a different value
- one pivot table at a time, to stay sane

build one on a column you assume is clean. watch what happens.

one agency, two rows

	A	B
1	SERVICETYPECODEDESCRIPTION	COUNTA of SERVICEREQUESTID
2	SWMA- Solid Waste Management Admistration	3199
3	PEMA- Parking Enforcement Management Administration	1916
4	SWMA- Solid Waste Management Administration	1199
5	311- Call Center	719
6	Toa-Street & Bridge Maintenance	544
7	(blank)	447
8	DOH- Department Of Health	314
9	Urban Forrestry	229
10	Toa- Trans Sys Mnt	201
11	Transportation Operations Administration	122
12	Grand Total	9153

Pivot table 1

THE FINDING

A single typo -
Admistration - splits Solid
Waste into two categories.

Filter for the 'correct' spelling
and you drop **3,199 of 4,398**
requests (73%).

*Don't assume that typos are a
single mistake.*

how many are “closed”?

	A	B
1	SERVICEORDERSTATUS	COUNTA of SERVICEREQUESTID
2	Canceled	53
3	Closed	8494
4	Closed (Duplicate)	156
5	Closed (Transferred)	9
6	In-Progress	358
7	Open	80
8	Open (Duplicate)	3
9	Grand Total	9153

Pivot table 1

THE FINDING

“Closed” isn't one number:
 $8,494 + 156 + 9 = 8,659$

Count only the bare ‘Closed’
and you under-report closed
orders by 165.

problems with names

- one agency, many aliases — DDOT files as:
 - Toa-Street & Bridge Maintenance · Toa- Trans Sys Mnt · Transportation Operations Administration · Department of Transportation
- misspellings quietly split categories: Admistration, Urban Forrestry
- the fix isn't only cleaning; you need to decide what counts as the same thing and document why

same trap, other datasets: AT&T vs AT and T, Jr./Sr., names that change with marriage.

problems with addresses

- DC addresses carry a quadrant, unless they don't
- 25 rows have no quadrant at all:
 - parks (UPSHUR PARK), circles & squares (THOMAS CIRCLE), block ranges, an I-295 exit ramp
- some aren't errors - they're places you can't easily geocode like a street address.

NW 4,274	NE 2,557
SW 222	SE 2,070

requests per quadrant

absence is a finding too

- some columns are 100% empty – INSPECTORNAME, INSPECTIONDATE, STATUS_CODE, DETAILS.
- RESOLUTIONDATE is blank on 441 requests - opened and, on paper, never closed. who, and where?
- the inconsistency might be the important bit!